# Sustainable Building Blocks to Produce Carotenoids

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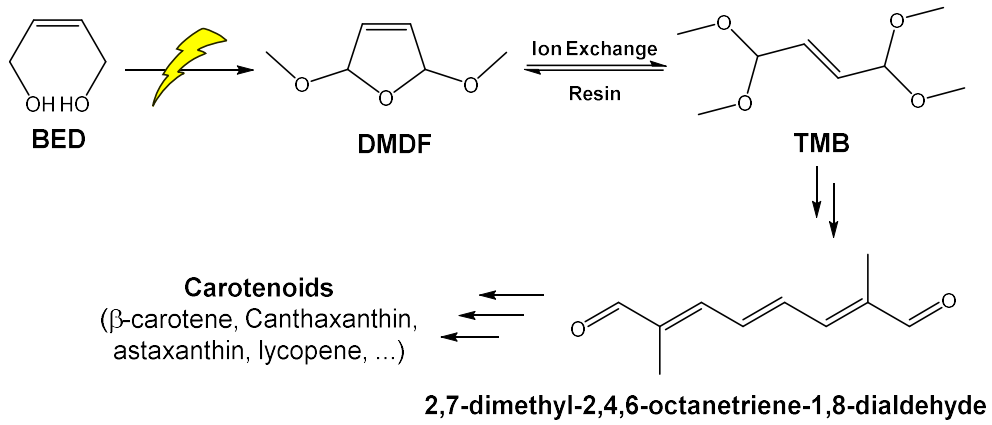
Carotenoids are a group of natural pigments that are vital for human and animal nutrition and health *e.g.*, as source of vitamin A. In addition, carotenoids add color to foods and beverages. The industrial manufacturing of most of the carotenoids is executed through chemical synthesis to manage the need and market demands. A key intermediate for carotenoids synthesis is 2,7-dimethyl-2,4,6-octanetriene-1,8-dialdehyde (see Scheme). [1] An important intermediate is 1,1,4,4-tetramethoxy-2-butene (TMB), which can be synthesized by acid catalyzed ring-opening reaction from 2,5-dimethoxydihydrofuran (DMDF). The current preparation method for TMB and DMDF faces challenges in enhancing reaction performance, managing waste, and avoiding the use of hazardous reagents.




Here we present results of the electrochemical oxidation of *Z*-2-butene-1,4-diol (BED) to DMDF, [2] optimized and scaled-up to 100 $cm^2$ electrode surface area. A two-stage nanofiltration for >99% recycling of the spent electrolyte and pervaporation for water removal were identified as the best options for further processing of the crude DMDF. [3] A complete process concept was developed. Purified DMDF was successfully converted in presence of an acidic ion exchange resin to TMB in a continuous mode resulting in an excellent selectivity control, while salt waste could be minimized. [4] Unconverted DMDF could easily be recycled and re-used. The new process fulfills the criteria for a green and sustainable production and engineering for

the carotenoid building block TMB. The downstreaming produces only minimal amount of wastewater (released during reaction). An LCA showed the significantly improved carbon food print for the new process by a potential decrease of >66% compared to the current process.


[1] *Carotenoids Vol.2: Synthesis*, Birkhäuser Verlag, Basel, Boston, Berlin, p. 117/118, 1996

[2] W. Bonrath, E. L. V. Goetheer, R. Goy, M. Joray, R. Latsuzbaia, Process for the preparation of alkoxylated 2,5-dihydrofuran, WO2023242064A1

[3] W. Bonrath, R. Goy, M. Jory, R. Wächter, Process for the preparation of 1,1,4,4-tetraalkoxy-2-butene, Patent filed

[4] W. Bonrath, R. Goy, M. Joray, Sustainable process for the manufacture of 1,1,4,4-tetramethoxy-2-butene, WO2021170864, 2021.